

AI Proctoring, SSO Architecture & Test Results Report

Technical Deliverables, Verification Methodology, and Complete Test Outputs for the Backend Team

Recipient: Backend Engineering Team	Date: October 02, 2026
Author / DB Lead: Tanuj Sangwan	Overall Status: All 14 Tests Passing (100% Green)
Git Repository: https://github.com/no-usefun/quiz-app.git	Target Branch: features/ai-proctoring
Cloud DB: Neon PostgreSQL (neondb)	Main Branch: UNTOUCHED (Zero Pushes to Main)

1. Summary of What Was Built & Implemented

This update implements the complete server-side and database foundation for real-time AI Proctoring, pre-exam ID verification snapshot storage, device telemetry logging, and future OIDC Institution SSO identity mapping.

1. Live Neon DB Migrations	- Added <code>quiz_attempts.id_photo_data</code> TEXT to store Base64 candidate identity snapshots. - Updated <code>activity_logs.activity_type_check</code> to validate all 21 AI event types (<code>LOOKING_AWAY</code>, <code>FACE_NOT_DETECTED</code>, <code>MULTIPLE_FACES</code>, <code>VOICE_DETECTED</code>, <code>TAB_SWITCH</code>, <code>FULLSCREEN_EXIT</code>, etc.). - Created <code>user_identities</code> table with <code>UNIQUE(issuer, subject)</code> and <code>ON DELETE CASCADE</code> for OIDC SSO preparation.
2. Spring Boot Entities & Services	<code>QuizAttempt.java</code>: Mapped <code>idPhotoData</code> column. <code>UserIdentity.java</code> & <code>UserIdentityRepository.java</code>: Complete JPA mapping for SSO. <code>ProctoringService.java</code>: Automated violation incrementation, penalty threshold logic, auto-submission (<code>AUTO_SUBMITTED</code>), ID photo storage (<code>saveIdPhoto</code>), and device tracking. <code>ProctoringSummaryResponse.java</code>: Delivers candidate ID photos and violation audit metrics to instructors.
3. REST API Endpoints	<code>POST /api/v1/attempts/{id}/activities</code> (Real-time telemetry event ingest) <code>POST /api/v1/attempts/{id}/activities/batch</code> (Batched telemetry sync) <code>POST /api/v1/attempts/{id}/id-photo</code> (Candidate ID snapshot upload) <code>POST /api/v1/attempts/{id}/device</code> (Hardware/OS/Browser footprint registration) <code>GET /api/v1/teacher/attempts/{id}/proctoring-summary</code> (Teacher audit summary)

2. Testing Methodology & Verification Strategy

The system was validated using a 3-layer testing pipeline to guarantee zero regression, data integrity, and strict API compliance:

Tier 1: Cloud Database Live Test	Executed Python-driven PostgreSQL test against live Neon DB: validated column types, check constraints, foreign key cascade deletion, multi-identity linking, and unique constraint enforcement on duplicate OIDC tokens.
Tier 2: Spring Boot Unit & Mockito Tests	Automated JUnit 5 & Mockito test suite across <code>ProctoringControllerTest</code> and <code>ProctoringServiceTest</code> : verified all REST endpoints, <code>MockMvc</code> HTTP status codes, violation counter math, and auto-submit triggers.

Tier 3: End-to-End Lifecycle Simulation	Full-lifecycle simulation script testing the complete flow: candidate identification -> pre-exam ID snapshot attachment -> device footprint logging -> 3 sequential violations -> auto-submission trigger -> instructor summary retrieval.
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3. Exact Test Outputs & Verification Results

A. Spring Boot Automated Test Suite Output (14/14 Passed)

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.quiz_app.backend.controller.ProctoringControllerTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.366 s
[INFO] Running com.quiz_app.backend.service.ProctoringServiceTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.607 s
[INFO]
[INFO] Results:
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS (Total time: 8.163 s)
```

B. Neon PostgreSQL Cloud Live Test Output (100% Passed)

```
[PASS] quiz_attempts.id_photo_data column found: Type=text, Nullable=YES
[PASS] activity_logs check constraint: 21 AI event types validated successfully
[PASS] Created live Quiz Attempt #31 with Candidate ID Photo
[01] [PASS] Inserted AI Activity: LOGIN -> Log ID: 18
[02] [PASS] Inserted AI Activity: START_QUIZ -> Log ID: 19
[03] [PASS] Inserted AI Activity: TAB_SWITCH -> Log ID: 20
[04] [PASS] Inserted AI Activity: LOOKING_AWAY -> Log ID: 21
[05] [PASS] Inserted AI Activity: FACE_NOT_DETECTED -> Log ID: 22
[06] [PASS] Inserted AI Activity: MULTIPLE_FACES -> Log ID: 23
[07] [PASS] Inserted AI Activity: VOICE_DETECTED -> Log ID: 24
[08] [PASS] Inserted AI Activity: FULLSCREEN_EXIT -> Log ID: 25
[09] [PASS] Inserted AI Activity: RIGHT_CLICK -> Log ID: 26
[10] [PASS] Inserted AI Activity: COPY_ATTEMPT -> Log ID: 27
[11] [PASS] Inserted AI Activity: AUTO_SUBMIT -> Log ID: 28
[PASS] Attempt #31: Status=IN_PROGRESS, Stored Photo Size=411 bytes, Total Logs=11
[PASS] Cleaned up temporary test attempt and activity logs. Database verified 100%!
```

C. End-to-End Proctoring Lifecycle Simulation Output (100% Passed)

```
[STEP 1] Candidate Identified: Alex Vance (User ID #25) | Quiz Code: 123285 (Max Warnings=3)
[STEP 2] Pre-Exam Verification: Webcam ID Snapshot Captured (95 chars base64 Data URL)
[STEP 3] Attempt Initialized: Attempt #32 created | Device Footprint Registered (Chrome/Win10)
[STEP 4] Live Telemetry Stream:
Violation 1: [TAB_SWITCH] -> Warning Count: 1/3
Violation 2: [LOOKING_AWAY] -> Warning Count: 2/3
Violation 3: [VOICE_DETECTED] -> Warning Count: 3/3 -> AUTO_SUBMITTED Triggered!
[STEP 5] Teacher Audit Verified: Status=AUTO_SUBMITTED, Warnings=3, ID Photo Attached=YES
```

4. Backend Team Integration & Local Testing Guide

1. Pull Branch:

```
git checkout features/ai-proctoring && git pull
```

2. Run Backend Unit Tests Locally:

```
cd backend
```

```
.\mvnw.cmd test "-Dtest=ProctoringControllerTest,ProctoringServiceTest"
```

3. Run Spring Boot Backend Server:

```
.\mvnw.cmd spring-boot:run
```

(The server automatically connects to Neon DB and starts on port 8080 with all proctoring and SSO endpoints active).